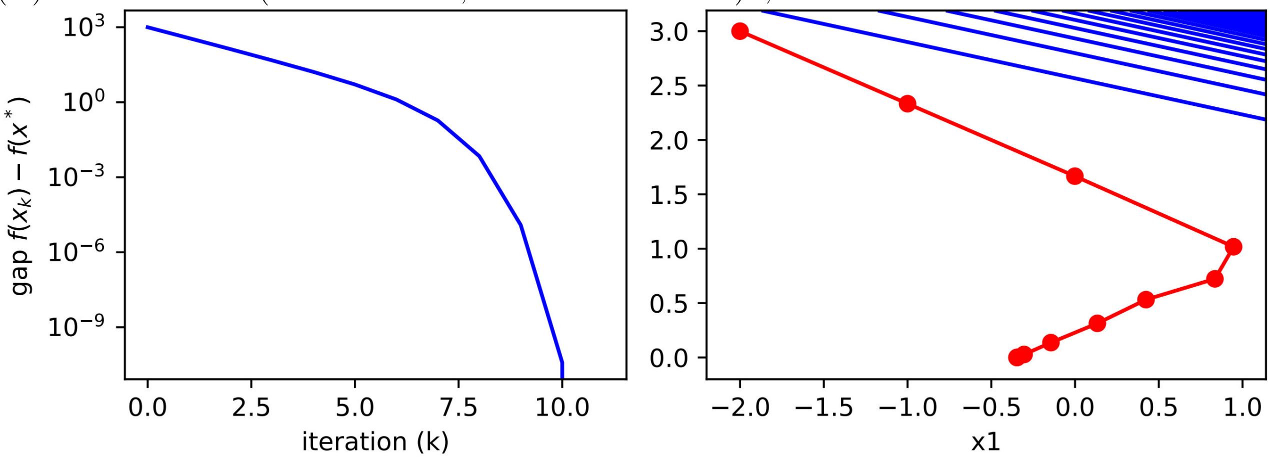
CS2601 Linear and Convex Optimization

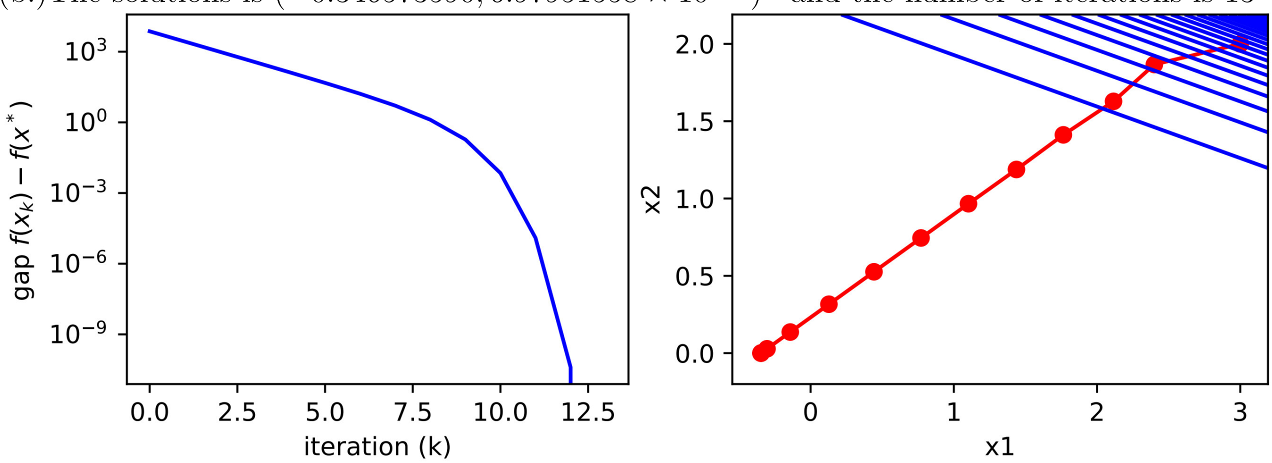
Homework 4

Name:Chen Zhuoyan   Student ID:523030910037

1. (a.)The solution is $(-0.346573590, 6.82106495 \times 10^{-12})^T$, and the number of iterations is 11

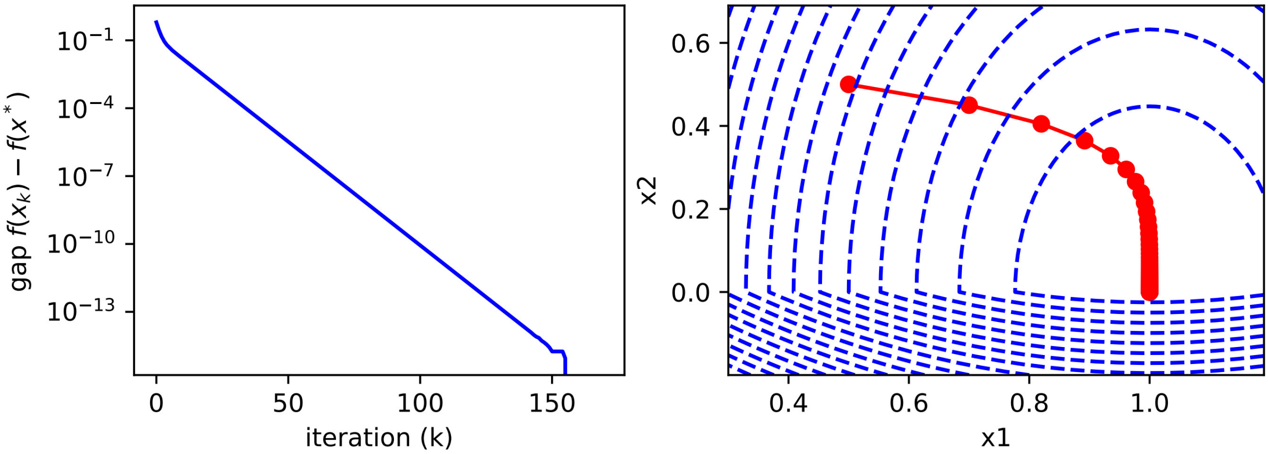


(b.)The solutions is $(-0.346573590, 6.97951558 \times 10^{-12})^T$ and the number of iterations is 13

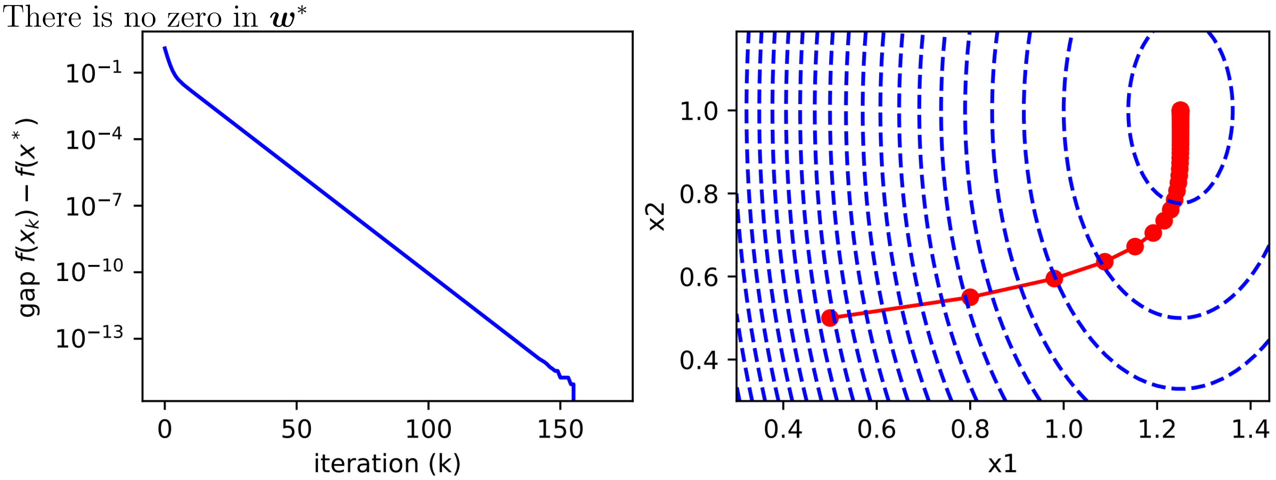


2. (a.)$f^{'}(x) = 4(x-a)^3, f^{''}(x) = 12(x-a)^2$
   Therefore the Newton's step is $x_{k+1} = x_k - \frac{f^{'}(x_k)}{f^{''}(x_k)} = \frac{2}{3}x_k + \frac{a}{3}$
   (b.)Because $x_{k+1} = \frac{2}{3}x_k + \frac{a}{3}$
   Therefore $y_{k+1} = |x_{k+1} - a| = |\frac{2}{3}x_k - \frac{2a}{3}| = \frac{2}{3}|x_k - a| = \frac{2}{3}y_k$
   (c.)Because $y_{k+1} = \frac{2}{3}y_k$
   Therefore $y_k = (\frac{2}{3})^{k-1}y_0 = (\frac{2}{3})^{k-1}|x_0 - a|, k \in N^+, lim_{k\to\infty}y_k = 0$
   Therefore $|x_k`a| = y_k$ decays to zero exponentially

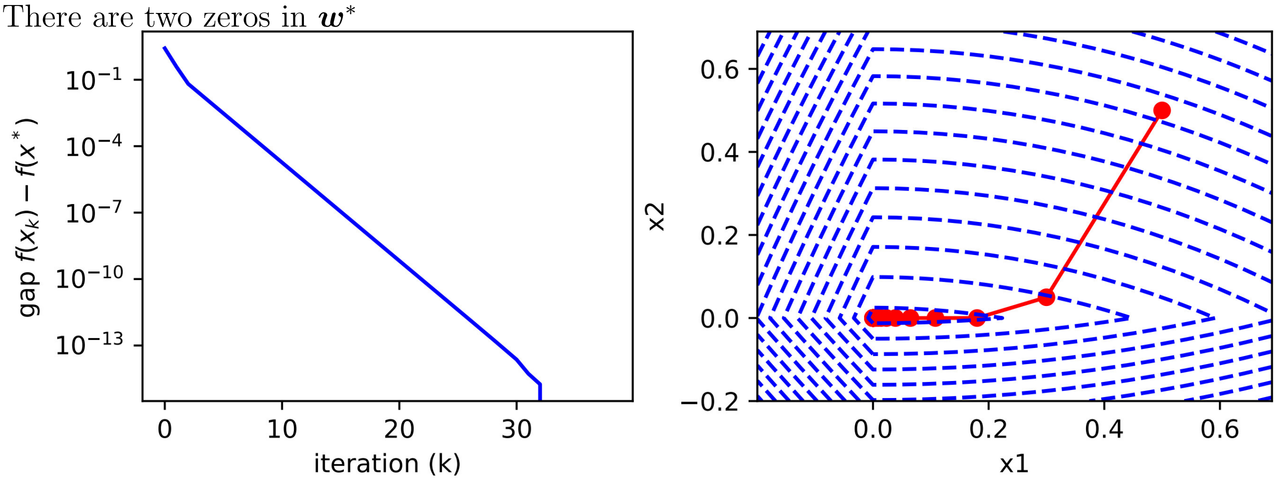
3. (a.)The solution is $(1.09.24600449 \times 10^{-9})^T$ and the number of iterations is 169



(b.)The solution is $(1.25, 0.99999999)^T$ and the number of iterations is 169.
There is no zero in $\boldsymbol{w}^*$



(c.)The solution is $(1.85659632 \times 10^{-9}, 0)^T$ and the number of iterations is 38
There are two zeros in $\boldsymbol{w}^*$



4. (a.)Because $x_1 + 2x_2 = 1$
   Therefore $x_1 = 1 - 2x_2, f(x_1, x_2) = g(x_2) = \frac{1}{2}(1-2x_2)^2 + x_2(1-2x_2) + x_2^2 - (1-2x_2) - 3x_2 = x_2^2 - 2x_2 - \frac{1}{2}$
   Let $g^{'}(x_2) = 2x_2 - 2 = 0, x_2 = 1$
   Therefore $x_1^* = -1, x_2^* = 1$
   (b.)$\mathcal{L}(x_1, x_2, \lambda) = \frac{1}{2}x_1^2 + x_1x_2 + x_2^2 - x_1 - 3x_2 + \lambda(x_1 + 2x_2 - 1)$

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x_1} = x_1 + x_2 - 1 + \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial x_2} = x_1 + 2x_2 - 3 + 2\lambda = 0 \\ x_1 + 2x_2 - 1 = 0 \end{cases}$$

Therefore $x_1^* = -1, x_2^* = 1, \lambda^* = 1$

5. (a.)$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{g}^T\boldsymbol{x} + c + \boldsymbol{\lambda}^T(\boldsymbol{Ax - b}), \boldsymbol{\lambda} \in R^k$
   The Lagrange condition is

$$\begin{cases} \nabla_{\boldsymbol{x}}\mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{Q}\boldsymbol{x}^* + \boldsymbol{g} + \boldsymbol{A}^T\boldsymbol{\lambda}^* = 0 \\ \nabla_{\boldsymbol{\lambda}}\mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{Ax^* - b} = 0 \end{cases}$$

   (b.)Because $\boldsymbol{Q} \succ \boldsymbol{O}$
   Therefore $\boldsymbol{Q}^{-1} \succ \boldsymbol{O}$
   Therefore $\forall \boldsymbol{x} \in R^k, \boldsymbol{A^T x} \in R^n, \boldsymbol{x}^T(\boldsymbol{AQ}^{-1}\boldsymbol{A}^T)\boldsymbol{x} = (\boldsymbol{A^T x})^T\boldsymbol{Q}^{-1}(\boldsymbol{A^T x}) > 0$
   Therefore $\boldsymbol{AQ}^{-1}\boldsymbol{A}^T \succ \boldsymbol{O}$, hence is invertible
   Therefore $\boldsymbol{\lambda}^* = -(\boldsymbol{AQ}^{-1}\boldsymbol{A}^T)^{-1}(\boldsymbol{b} + \boldsymbol{AQ}^{-1}\boldsymbol{g}), \boldsymbol{x}^* = -\boldsymbol{Q}^{-1}(\boldsymbol{g} - \boldsymbol{A}^T(\boldsymbol{AQ}^{-1}\boldsymbol{A}^T)^{-1}(\boldsymbol{b} + \boldsymbol{AQ}^{-1}\boldsymbol{g}))$
   (c.)$\frac{1}{2}||\boldsymbol{x} - \boldsymbol{x}_0||_2^2 = \frac{1}{2}(\boldsymbol{x}^T - \boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{x} - \boldsymbol{x}_0^T\boldsymbol{x} + \frac{1}{2}||\boldsymbol{x}_0||_2^2 = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{x}$ when $\boldsymbol{x}_0 = \boldsymbol{0}$
   From (b.) we can know that $\boldsymbol{A} = [1, 2], b = 1, \boldsymbol{x}^* = [\frac{1}{5}, \frac{2}{5}]^T, \boldsymbol{\lambda}^* = -\frac{1}{5}$
   (d.)$dist(\boldsymbol{x}_0, P) = min||\boldsymbol{x} - \boldsymbol{x}_0||, \boldsymbol{w}^T\boldsymbol{x} = b$
   From (c.) we can know that $min_{\boldsymbol{x}}||\boldsymbol{x} - \boldsymbol{x}_0|| = min_{\boldsymbol{x}}\frac{1}{2}||\boldsymbol{x} - \boldsymbol{x}_0||_2^2 = \frac{\boldsymbol{w}b}{||\boldsymbol{w}||^2}$
   $\therefore dist(\boldsymbol{x}_0, P) = \frac{||\boldsymbol{w}\boldsymbol{x}_0 + b||}{||\boldsymbol{w}||}$